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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference C to diameter d ?

- a. $d = C / \pi$
- b. $d = 2C / \pi$
- c. $d = C^2 / \pi$
- d. $d = \pi C$
- e. $d = \sqrt{(C/\pi)}$

Answer Q1 ☒ ☐ B ☐ C ☐ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Adjusted R^2
- b. R^2
- c. Root Mean Squared Error (RMSE)
- d. Correlation coefficient r
- e. p-value of the slope

Answer Q2 ☐ A ☐ B ☒ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- b. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
- c. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- d. Measure a few heights, take the mean, and assign it to all trees.
- e. Map crowns in an equal-area projection and use crown area as height proxies.

Answer Q3 ☒ ☐ B ☐ C ☐ D ☐ E

Q4. Why did the lecture emphasize using simple formulae carefully before moving into software tools?

- a. Software can only run if every formula is typed from memory.
- b. Quantitative interpretation still depends on understanding what measurements and formulae represent.
- c. Coding removes the need to think about units or measurement.
- d. Environmental science avoids mathematical calculations once data are collected.
- e. Formulae are only relevant for drawing figures, not analysis.

Answer Q4 ☐ A ☒ B ☐ C ☐ D ☐ E



Q5. In the lecture, what was the purpose of discussing RStudio setup before the labs?

- a. To make sure students could use the required computing environment for practical work.
- b. To replace all later lectures with self-guided programming tasks.
- c. To discourage students from using local computers.
- d. To assess students before they had seen the course material.
- e. To demonstrate that coding is separate from environmental science.

Answer Q5 ☒ A ☐ B ☐ C ☐ D ☐ E

Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is the default tool for all students regardless of their computer setup.
- b. It is a backup option for students who cannot run the local software environment.
- c. It is used only for watching lecture videos.
- d. It automatically calculates all answers without coding.
- e. It is required before students can learn RStudio.

Answer Q6 ☐ A ☒ B ☐ C ☐ D ☐ E



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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference C to diameter d ?

- a. $d = C^2 / \pi$
- b. $d = \sqrt{C/\pi}$
- c. $d = C / \pi$
- d. $d = \pi C$
- e. $d = 2C / \pi$

Answer Q1 ☐ A ☐ B ☒ C ☐ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Root Mean Squared Error (RMSE)
- b. p-value of the slope
- c. Adjusted R^2
- d. Correlation coefficient r
- e. R^2

Answer Q2 ☒ A ☐ B ☐ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- b. Map crowns in an equal-area projection and use crown area as height proxies.
- c. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- d. Measure a few heights, take the mean, and assign it to all trees.
- e. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.

Answer Q3 ☐ A ☐ B ☒ C ☐ D ☐ E

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- e. To replace all later lectures with self-guided programming tasks.

Answer Q5 ☐ A ☐ B ☒ C ☐ D ☐ E

Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is used only for watching lecture videos.
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Answer Q6 ☐ A ☐ B ☐ C ☐ D ☒ E



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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height–DBH model. What is the correct conversion from measured circumference C to diameter d ?

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- b. $d = C^2 / \pi$
- c. $d = 2C / \pi$
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- e. $d = \pi C$

Answer Q1 ☐ A ☐ B ☐ C ☒ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. p-value of the slope
- b. Root Mean Squared Error (RMSE)
- c. R^2
- d. Adjusted R^2
- e. Correlation coefficient r

Answer Q2 ☐ A ☒ B ☐ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Map crowns in an equal-area projection and use crown area as height proxies.
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- d. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- e. Measure a few heights, take the mean, and assign it to all trees.

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Answer Q5 ☐ A ☐ B ☐ C ☒ D ☐ E

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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference C to diameter d ?

- a. $d = 2C / \pi$
- b. $d = \pi C$
- c. $d = \sqrt{C/\pi}$
- d. $d = C^2 / \pi$
- e. $d = C / \pi$

Answer Q1 ☐ A ☐ B ☐ C ☐ D ☒

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. R^2
- b. Correlation coefficient r
- c. p-value of the slope
- d. Root Mean Squared Error (RMSE)
- e. Adjusted R^2

Answer Q2 ☒ A ☐ B ☐ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
- b. Measure a few heights, take the mean, and assign it to all trees.
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- d. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
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Answer Q3 ☐ A ☐ B ☐ C ☐ D ☒

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Answer Q4 ☐ A ☐ B ☐ C ☒ D ☐ E



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- b. To assess students before they had seen the course material.
- c. To demonstrate that coding is separate from environmental science.
- d. To discourage students from using local computers.
- e. To make sure students could use the required computing environment for practical work.

Answer Q5 ☐ A ☐ B ☐ C ☐ D ☒ E

Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is a backup option for students who cannot run the local software environment.
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Answer Q6 ☐ A ☐ B ☐ C ☒ D ☐ E



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- a. $d = C / \pi$
- b. $d = 2C / \pi$
- c. $d = C^2 / \pi$
- d. $d = \pi C$
- e. $d = \sqrt{C/\pi}$

Answer Q1 ☐ A ☒ B ☐ C ☐ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Adjusted R^2
- b. R^2
- c. Root Mean Squared Error (RMSE)
- d. Correlation coefficient r
- e. p-value of the slope

Answer Q2 ☐ A ☐ B ☐ C ☒ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- b. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
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Answer Q4 ☒ A ☐ B ☐ C ☐ D ☐ E



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- e. To demonstrate that coding is separate from environmental science.

Answer Q5



Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is the default tool for all students regardless of their computer setup.
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Answer Q6



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- e. $d = 2C / \pi$

Answer Q1 ☐ A ☐ B ☐ C ☒ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Root Mean Squared Error (RMSE)
- b. p-value of the slope
- c. Adjusted R^2
- d. Correlation coefficient r
- e. R^2

Answer Q2 ☐ A ☒ B ☐ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- b. Map crowns in an equal-area projection and use crown area as height proxies.
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- d. Measure a few heights, take the mean, and assign it to all trees.
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Answer Q3 ☒ A ☐ B ☐ C ☐ D ☐ E

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Answer Q4 ☐ A ☐ B ☒ C ☐ D ☐ E



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- e. To replace all later lectures with self-guided programming tasks.

Answer Q5 ☐ A ☐ B ☐ C ☐ D ☒ E

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Answer Q6 ☐ A ☐ B ☐ C ☒ D ☐ E



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- e. $d = \pi C$

Answer Q1 (A) (B) (C) (D) ☒

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. p-value of the slope
- b. Root Mean Squared Error (RMSE)
- c. R^2
- d. Adjusted R^2
- e. Correlation coefficient r

Answer Q2 (A) (B) ☒ (D) (E)

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Map crowns in an equal-area projection and use crown area as height proxies.
- b. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
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Answer Q3 (A) ☒ (C) (D) (E)

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Answer Q4 (A) (B) (C) ☒ (E)



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Answer Q6 ☐ A ☐ B ☐ C ☐ D ☒



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- a. R^2
- b. Correlation coefficient r
- c. p-value of the slope
- d. Root Mean Squared Error (RMSE)
- e. Adjusted R^2

Answer Q2 ☐ A ☐ B ☐ C ☒ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
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- e. To make sure students could use the required computing environment for practical work.

Answer Q5 ☐ A ☐ B ☒ C ☐ D ☐ E

Q6. Which statement best describes the role of a cloud workaround such as Google Colab in the lecture?

- a. It is a backup option for students who cannot run the local software environment.
- b. It automatically calculates all answers without coding.
- c. It is required before students can learn RStudio.
- d. It is used only for watching lecture videos.
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Name _____

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5	☐	☐	☐	☐	☐	5
6	☐	☐	☐	☐	☐	6
7	☐	☐	☐	☐	☐	7
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Q1. In fieldwork you often tape the circumference of a tree at breast height but need diameter (DBH) to use a height-DBH model. What is the correct conversion from measured circumference C to diameter d ?

- a. $d = C / \pi$
- b. $d = 2C / \pi$
- c. $d = C^2 / \pi$
- d. $d = \pi C$
- e. $d = \sqrt{(C/\pi)}$

Answer Q1 ☐ A ☐ B ☒ C ☐ D ☐ E

Q2. You are comparing two models that predict a vegetation variable Y from a sensor index X . Your priority is to minimise absolute prediction error in interpretable units of Y . Which metric should guide your choice?

- a. Adjusted R^2
- b. R^2
- c. Root Mean Squared Error (RMSE)
- d. Correlation coefficient r
- e. p-value of the slope

Answer Q2 ☒ A ☐ B ☐ C ☐ D ☐ E

Q3. You can easily measure DBH but not height in a forest survey. Which workflow best matches the lecture's definition of a model and the purpose of prediction?

- a. Fit a statistical model of height from DBH using a training set, then use it to predict heights for the remaining trees.
- b. Use Pythagoras with GPS coordinates to compute tree height from planimetric distances.
- c. Assume a fixed height:diameter ratio from a paper and multiply all DBHs by that constant.
- d. Measure a few heights, take the mean, and assign it to all trees.
- e. Map crowns in an equal-area projection and use crown area as height proxies.

Answer Q3 ☐ A ☐ B ☐ C ☒ D ☐ E

Q4. Why did the lecture emphasize using simple formulae carefully before moving into software tools?

- a. Software can only run if every formula is typed from memory.
- b. Quantitative interpretation still depends on understanding what measurements and formulae represent.
- c. Coding removes the need to think about units or measurement.
- d. Environmental science avoids mathematical calculations once data are collected.
- e. Formulae are only relevant for drawing figures, not analysis.

Answer Q4 ☐ A ☐ B ☐ C ☐ D ☒ E



Q5. In the lecture, what was the purpose of discussing RStudio setup before the labs?

- a. To make sure students could use the required computing environment for practical work.
- b. To replace all later lectures with self-guided programming tasks.
- c. To discourage students from using local computers.
- d. To assess students before they had seen the course material.
- e. To demonstrate that coding is separate from environmental science.

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6	☐	☐	☐	☐	6	☐
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- c. Adjusted R^2
- d. Correlation coefficient r
- e. R^2

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